Rewind & Amplify

Predicting how hidden defects change quantum measurement patterns

A Qiskit model tested on IBM hardware.
Exploratory extension: laser diagnostics.

Qubits / Florida Gulf Coast University

Florida Qiskit Hackathon finals / FAU / 14 November 2026



Hidden defects can change the state before readout

State estimates depend on the dynamics and readout model

During qubit evolution

A coupled defect can exchange energy or change the qubit's frequency.

The intended state can evolve differently from the ideal circuit.

When interpreting the readout

Repeated measurements estimate probabilities, not a full state in one shot.

Readout errors and hidden dynamics can otherwise receive the wrong explanation.

Potential value: better calibration and more trustworthy state estimates

State estimation needs appropriate measurements and a validated noise model. This project has not diagnosed a physical defect.

A click is one recorded measurement outcome

1 = click

The output qubit reads 1 in one model time step.

0 = no click

A single click does not identify a defect.

Time step	1	2	3	4
One example record	0	0	1	0

The emitter and hidden qubit carry memory between steps

Optical interpretation: a detector fires. On IBM, superconducting qubits represent this model.

The QRSM connects each bit to a hidden-state model

Does an emitter alone explain the record, or does a hidden partner help?

Updates the modeled state

Each observed 0 or 1 updates the conditional state of the emitter and its hidden partner.

Scores a physical explanation

Candidate coupling and noise settings assign a probability to the record and predict the next readout.

A fresh synthetic test: detector dead time changes the inferred coupling

True coupling	Dead time ignored	Dead time modeled
0.65	0.069	0.706

Other fitted parameters remained biased. Classical sequence models also use history. IBM used known models.

Hidden memory changes what follows a click

Given a click at step 3, how likely is another click later?



At step 6

10.10%

without hidden defect

4.67%

with hidden defect

Exact model predictions. We compare known model alternatives. No unknown physical defect was diagnosed.

Rare conditions waste ordinary samples

3.95%

of ideal runs have the selected click

About 25 runs per matching sample

Quantum query scaling

$1 / p$

ordinary rejection

$1 / \sqrt{p}$

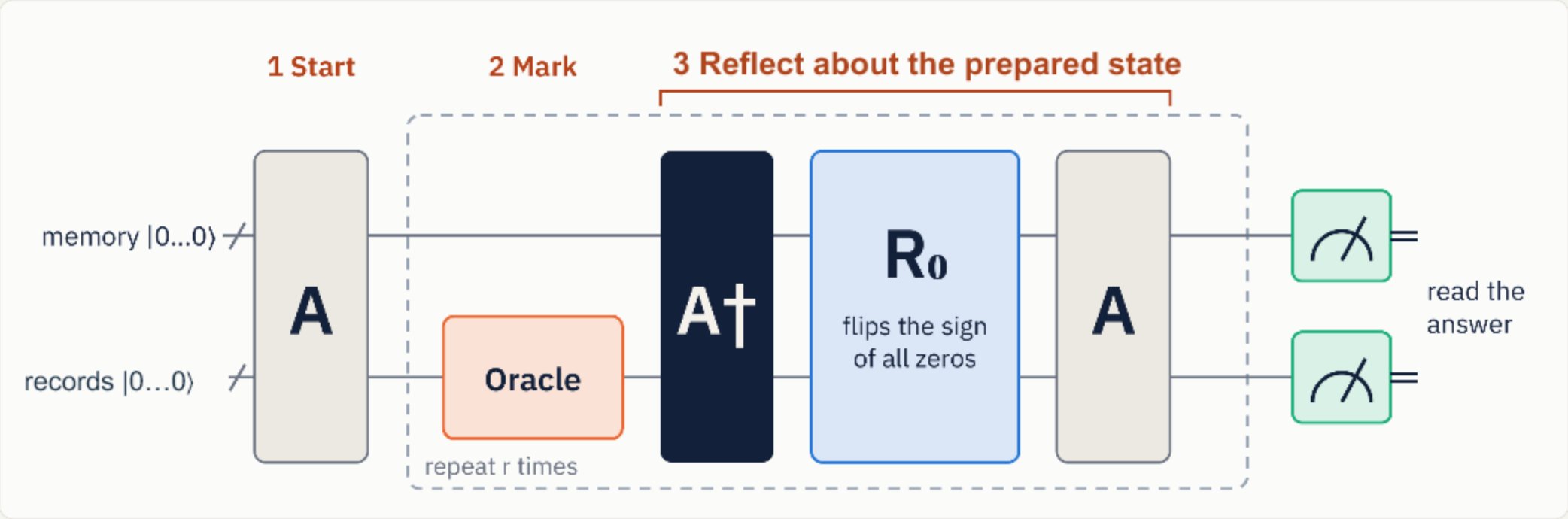
ideal coherent model calls

p = probability of the selected event

Today, a classical conditional sampler wins this small problem

A possible benefit needs larger quantum memory and sufficiently accurate reversible circuits.

A reversible model makes amplification possible



Save the history

Fresh qubits
replace resets

Steer the model

Classical tuning
of a supported event

Amplify locally

Fewer gates
with approximation

Check the answer

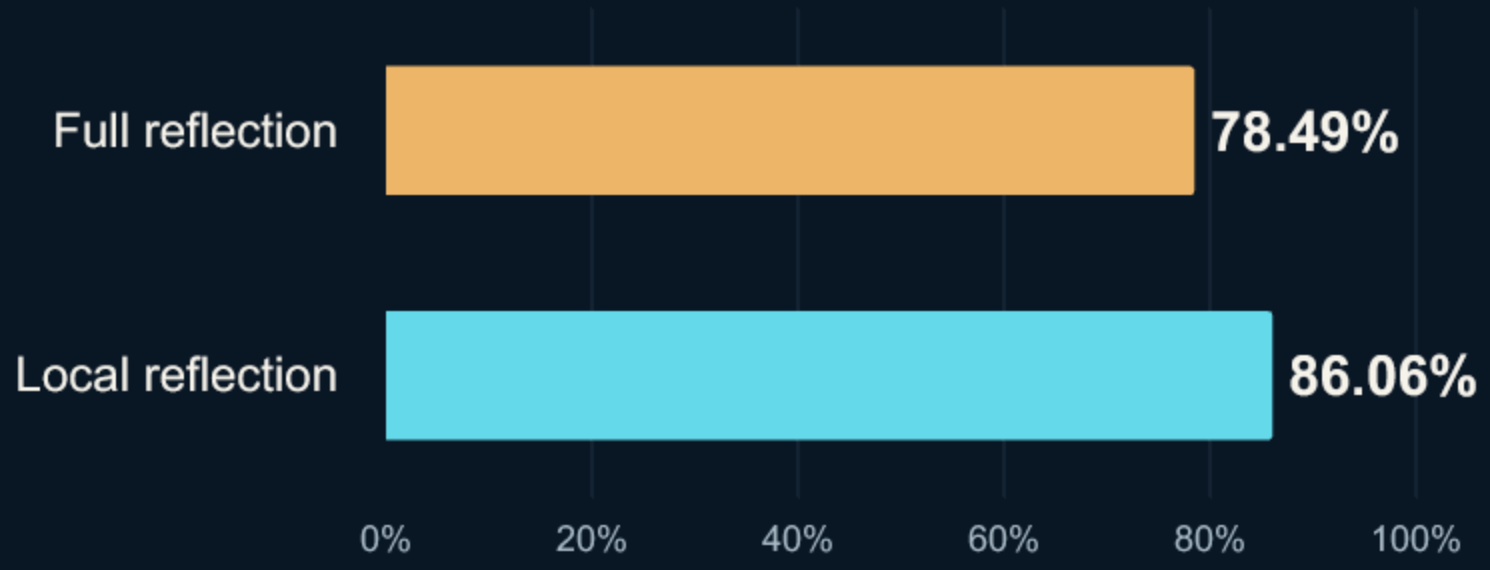
Event rate and
conditional accuracy

Qiskit circuit construction, Aer verification, IBM compilation and archived device counts.

More matching answers with fewer gates

Actual IBM Kingston run. Same steering, one amplification round.

Raw matching-event rate



40 vs 85

two-qubit gates
local / full

0.055 / 0.069

conditional-history error*
local / full

Cost matters: fewer executed gates per accepted answer in this pair

4,096 shots each. *Corrected TVD, lower is closer. Layouts differ. Gate accounting excludes other costs.

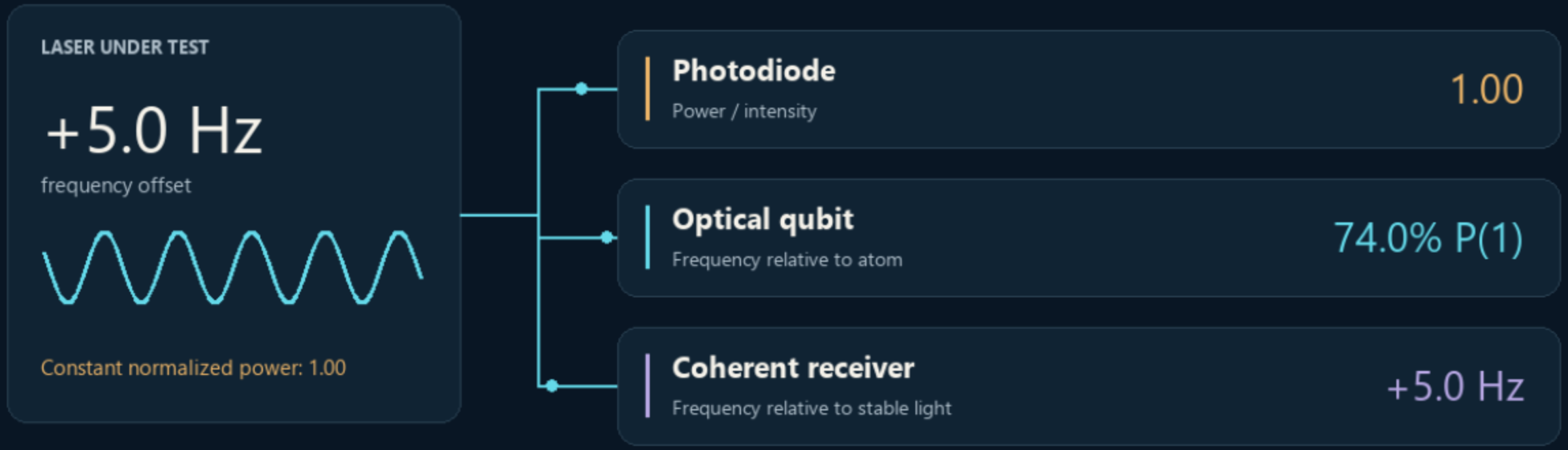
The query returns a prediction about the next clicks



Live local demonstration of the original model. Static screenshot remains available if switching fails.

Laser diagnostics could protect chip exposures

Exposure light affects the pattern printed on a wafer

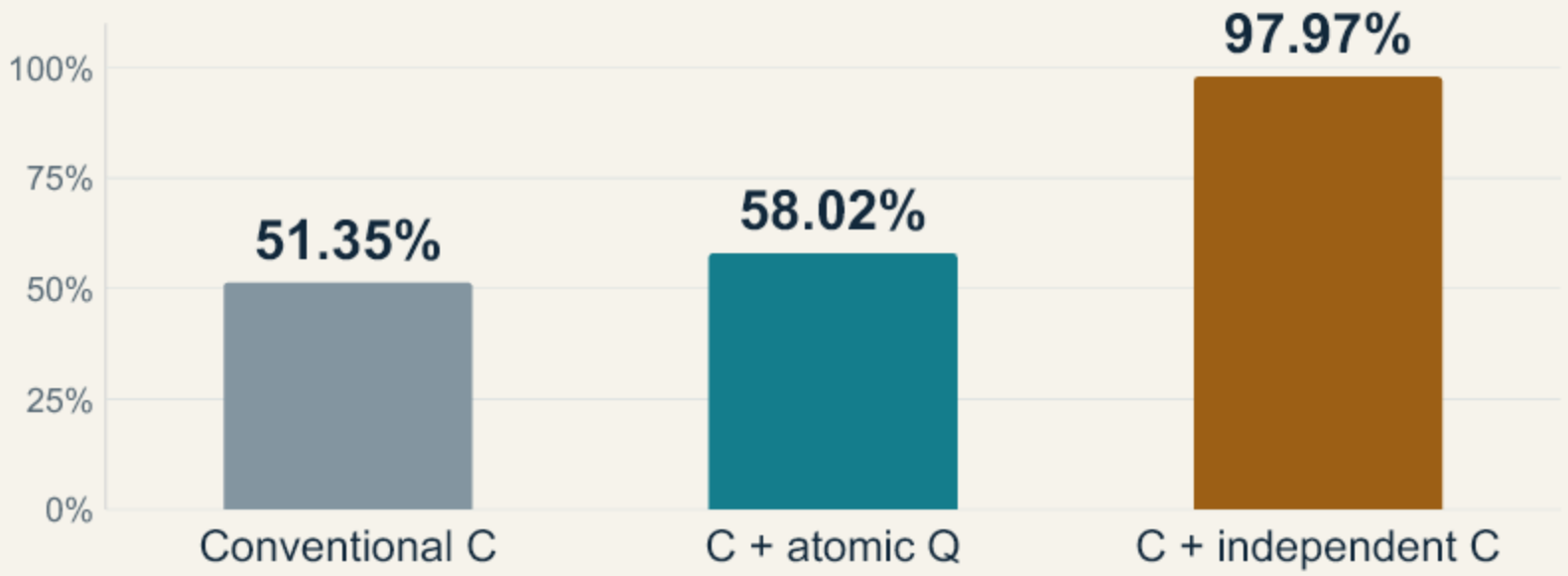


Research question: does an atomic reference add useful information?

Illustrative +5 Hz response. Our 698 nm lab surrogate needs a separate sensing design for lithography sources.

The classical reference leads with unknown onset

Detection within 32 s of a ±0.5 Hz step. Its start time is hidden.



Atomic addition

+6.67 points

95% interval
+3.96 to +9.38 points

C + C still leads

Full-record false alarms: 5.42% / 5.16% / 4.06% (C / C + Q / C + C)

128 s records. 1,920 per class. Unseen sessions. Nominal 5% alarm target. Total apparatus costs differ.

The evidence changes our next experiment

Drive-amplitude monitoring

3.71% timely detection

6.84% false source alarms

This budget gives no useful prompt warning

Revise the operating point before a sensor prototype.

Phase inspection model

0.00637 vs 0.00791 rad

Hybrid / coherent phase error

A conditional precision benefit merits a calibrated optical test

20,000 incident photons
81% overall efficiency

Separate simulations. Drive: +20% field step, 32-bin detection horizon, 96-bin alarm record. No wafer result.

Our contribution and what we learned

A queryable recurrent quantum model

Reversible construction and hardware adaptations,
with exact-model checks and archived IBM results

Controls changed our conclusions. Accuracy and total cost both matter.

Nicholas Taylor

QRSM research, Grover fusion, experiments

Timofey Yudintsev

Research, keyword-search idea

Dan Izquierdo

Experimentation

Dillon Satria Sitepu

Formatting and revising

QRSM model predates the event. Known amplification methods. AI assistance: Claude and Codex / OpenAI.

Hardware evidence and ablations

Circuit	Raw hits	Corrected hits	2-qubit gates	TVD*
Plain sampling	6.05%	5.98%	13	0.071
Steering only	21.36%	21.43%	13	0.090
Steer + full, 1 round	78.49%	79.29%	85	0.069
Steer + local, 1 round	86.06%	87.36%	40	0.055
Full, unsteered, 3 rounds	58.52%	58.95%	213	0.263

Repeat: local 87.26% raw, full 78.61% raw

4,096 shots/circuit in each job. *Readout-corrected conditional-history TVD. Layouts differ.

Three optical bins expose the circuit cost

All three bins must click. Full reflection preserves the retained model.

Choi cutoff	Qubits	Whole-query CX	Accepted rate[1]	State error*
0.03	15	10,189	0.12783%	0.02604
0.003	18	19,864	0.12755%	0.01094

Implementation error < 6.3×10^{-13}

Exact classical recursion already answers these small queries efficiently

[1]After one round. *Trace distance versus the cavity-2 model. No three-bin noise, routing or physical sensor test.

Sensor assumptions and resource accounting

Model element	Assumed setting
Compatible laboratory wavelength	698 nm, narrow-line optical drive
Atomic ensemble	64 independent, unentangled atoms
Interrogation	2.5 ms pulse + 20 ms dark time + 2.5 ms pulse
Record and acquisition gap	128 s record, 975 ms dead time per cycle
Relaxation and readout	T_2 = 200 ms, T_1 = 100 s; 2% / 4% readout errors

Time and task match. Total apparatus resources do not.

Photon allocation is assumed bookkeeping. Reference lasers, cooling, trapping and readout costs remain uncounted.

Calibration failures can reverse the conclusion

Scenario	C detection / false alarm	C + Q	C + C
Unseen nominal sessions	51.4% / 5.4%	58.0% / 5.2%	98.0% / 4.1%
Shorter atomic coherence	52.8% / 6.0%	47.3% / 6.1%	98.1% / 5.0%
Primary-reference step	44.0% / 91.5%	45.6% / 84.5%	97.7% / 4.8%
Shared-path step	45.6% / 91.1%	48.2% / 94.1%	56.1% / 100.0%

Shared-path drift is indistinguishable from source drift here

Each cell: detection within 32 s / false alarms per 128 s record. 1,920 null and source records per scenario.

Detector dead time can distort inferred coupling

Fresh synthetic records. True coupling = 0.65. Same data for both fits.

Physical model	Fitted coupling	Held-out score relative to HMM*
Dead time ignored	0.069	−0.4506 nats / record
Blind-bin dynamics included	0.706	+0.0078 nats / record

Modeling dead time improves prediction by 0.4584 nats / record

Corrected parameters still show bias and a dephasing boundary

*24-bin records. Corrected model versus HMM interval spans zero. Synthetic inference, not physical defect diagnosis.

Where the quantum component enters

Layer	What it does	Evidence today
Sensor physics	Atomic state responds to sampled laser	Classical Bloch simulation
Diagnostic inference	Combines measured or simulated records	Classical statistical estimator
Rare-history query	Amplifies a condition in a coherent model	Qiskit simulation + IBM surrogate

A sensing benefit does not establish a computing speedup

The current sensor benchmark is not compiled into the IBM circuit or the small optical queries.

The transfer to lithography needs its own test

Signal	Physical role	Current model
193 nm excimer light	DUV exposure source	Not represented
CO_2 drive laser	Creates a tin plasma for EUV	Not represented
13.5 nm EUV	Plasma emission used for exposure	Not represented
698 nm laboratory laser	Compatible narrow-line atomic probe	Synthetic surrogate

Conventional source diagnostics already exist

Proposed value: test complementary information on synchronized observations, then assess a specific source.

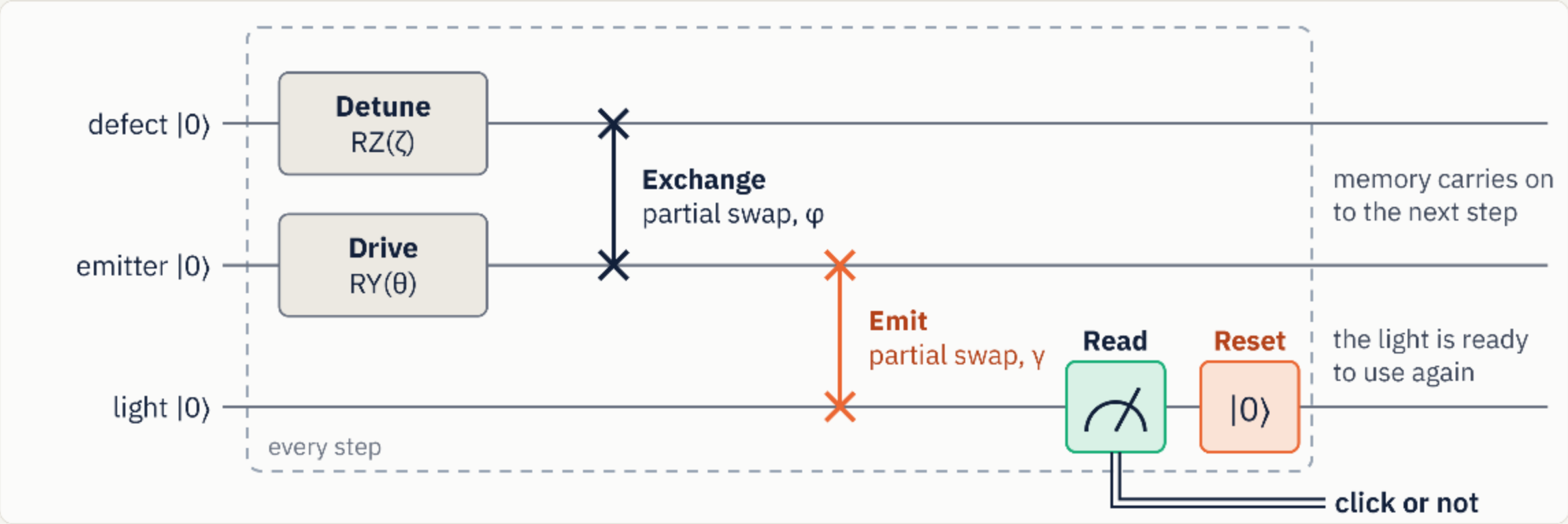
The continued query also ran on IBM

Model and circuit	Accepted / 4,096	Click at step 6*	2-qubit gates
Hidden defect, plain	290	12.07%	34
Hidden defect, steer + local	3354	6.89%	65
No defect, plain	503	13.72%	18
No defect, steer + local	3748	10.57%	29

Exact targets: 4.67% with defect; 10.10% without

*Raw conditional rate given step-3 click. Separate, longer circuits than the main 86.06% result.

One step of the recurrent quantum model



Dynamic version: measure and reset the output

Purified version retains a fresh output for each step and defers measurement, preserving the click statistics.

Judging rubric and evidence map

Criterion	Weight	Where the talk addresses it
Problem and impact	15%	1–5, 8, 10, 12: purpose, measurement, cost
Quantum rationale	20%	4, 6–7, 15, 27: model, scaling, limits
Technical Qiskit implementation	20%	7–9, 14–15, 21–22, 27: circuit evidence
Results and evaluation	10%	4–5, 8, 11–12, 17–18, 25–29
Innovation	10%	7, 13, 24: contribution versus prior work
Presentation	15%	13 main slides, planned 9:20 including demo
Q&A and teamwork	10%	13–29: roles, lessons, evidence backups

Hardware bonus: highest applicable tier, up to +5. Evidence supports evaluation, not a guaranteed score.

Contribution, prior work and AI assistance

Origin	What belongs there
Established prior work	Amplitude amplification and amplified conditional inference
Before the event	QRSM idea, physics-shaped model and exact filter
Original hackathon work	Qiskit realization, Grover fusion, steering, local reflection, experiments
Finals extension	Independent model audits, laser diagnostics and optical feasibility studies
AI assistance	Original: Claude. Finals: Codex with OpenAI models

A sensing benefit and a computing speedup are separate claims

Atomic laser-noise spectroscopy also predates this project. No new atomic sensing principle is claimed.

Drive amplitude: a useful negative result

Same 96-bin records. Unknown +20% field step. Detection within 32 bins.

Detector	Timely detection	Stable-record false alarms
Physical QRSM	3.71%	6.84%
Classical HMM	3.13%	5.27%
Count baseline	1.76%	1.95%

Prior-only RMSE 0.1136 → observed-data RMSE 0.1125

Most of the apparent estimation gain came from the prior

Post-holdout prior control is descriptive. 512 records per case; 6,656 total. No physical time or photon-budget calibration.

Phase inspection: benefit depends on the budget

Two illuminated sample modes. All incident photons are charged before loss.

Protocol at 81% efficiency	Full 2π phase RMSE	Wrong π branch
Coherent reference	0.00791 rad	0 / 2,000 trials
Pure two-photon NOON	2.22674 rad	50.4%
10% coherent + 90% NOON	0.00637 rad	0 / 2,000 trials

20,000 mean incident photons; ideal preparation in this comparison

1% preparation: matched dose takes 100× longer for pure NOON

Synthetic phase task, known reference and loss. Hybrid pilots are charged. Zero observed branch errors is not a guarantee.

The full spatial query is costly to construct

49 exposure cells, capped photon counts and a reversible weighted threshold

Constructed bound	Qubits	Generic CX upper bound	Dose-1.5 event probability
Floor weights	389	3,382,108	0.00423713
Ceiling weights	389	2,874,028	0.00422695

196 count qubits + 23 score qubits + 170 extra clean ancillas

Both circuit files round-trip. Classical evaluation is already tractable.

Constructive, unoptimized bounds. No full-state simulation, routing, hardware execution or physical optical claim.

Exposure: contrast trades against dose and loss

Phenomenological two-photon resist; fixed threshold and 20 evaluated guard cells

Illumination	Best mean cell accuracy	Incident photons	All 20 cells correct
Two-photon NOON	95.16%	14.88 million	36.76%
Phase-cycled coherent	78.28%	6.81 million	0.73%
10% duty coherent	83.29%	0.845 million	2.52%

Quantum contrast helps this chosen response; it is not dose advantage

Loss, blur and linear background can erase the pattern benefit

Each row uses its own best tested dose. All phase-cycle exposures charged. Higher classical peak intensity is allowed.

Model accuracy is part of the resource budget

Cavity truncation changes conditional predictions, even with exact circuit execution

Cavity levels	Default drive error*	Higher-drive error*
2	2.7643 percentage points	7.0829 percentage points
4	0.0494 percentage points	0.1118 percentage points
5	0.00281 percentage points	0.01102 percentage points

Controlled records improve some model comparisons; ambiguity remains

Classical filtering remains the baseline. More physics costs more qubits.

*Largest tested after-click probability error versus cavity level 6. Finite truncation study, not universal convergence.